

Week18: Pertussis and the CMI-PB project

Candice Lei (A18585167)

2026-06-01

Table of contents

Investigating Pertussis Cases by Year	1
A Tale of Two Vaccines (wP & aP)	2
Exploring CMI-PB Data	4
Examine IgG Ab Titer Levels	11
Obtaining CMI-PB RNASeq Data	18

Investigating Pertussis Cases by Year

Q1. With the help of the R “addin” package [datapasta](#) assign the CDC pertussis case number data to a data frame called `cdc` and use **ggplot** to make a plot of cases numbers over time.

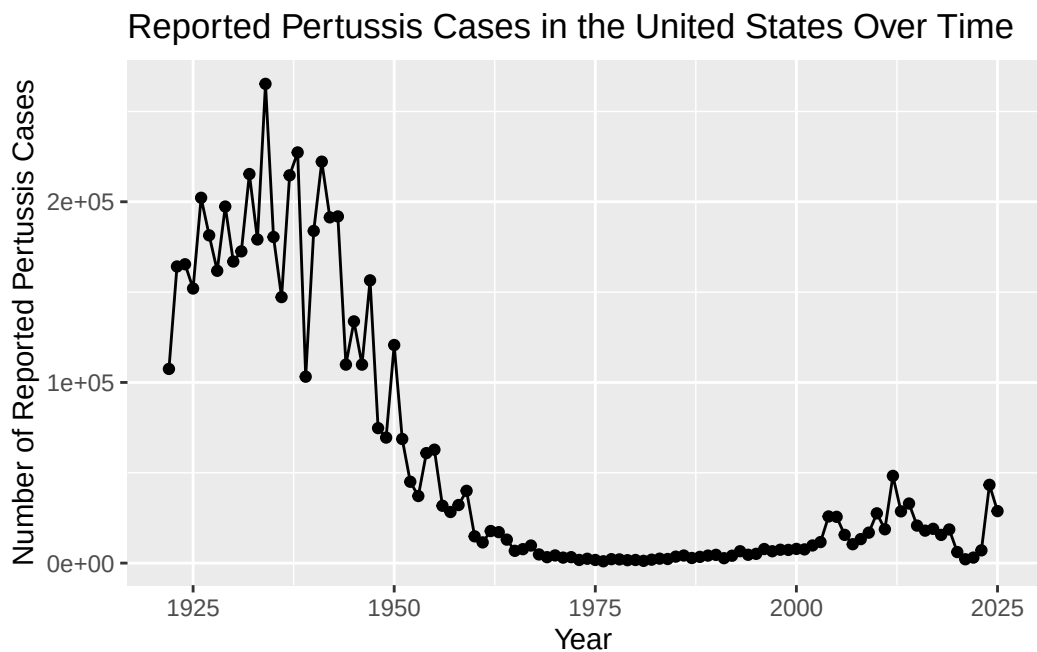
```
library(ggplot2)
cdc <- read.csv("~/Downloads/U.S. Reported Pertussis Cases_ 1922 - 2025.csv")
names(cdc)
```

```
[1] "Year"                                "Number.of.Reported.Pertussis.Cases"
[3] "Data.Status"
```

```
cdc$Number.of.Reported.Pertussis.Cases <- as.numeric(
  gsub(",", "", cdc$Number.of.Reported.Pertussis.Cases)
)

ggplot(cdc) +
  aes(x = Year, y = Number.of.Reported.Pertussis.Cases) +
```

```
geom_point() +
geom_line() +
labs(
  title = "Reported Pertussis Cases in the United States Over Time",
  x = "Year",
  y = "Number of Reported Pertussis Cases"
)
```



A Tale of Two Vaccines (wP & aP)

Q2. Using the ggplot `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

I notice that after wP vaccine was introduced in 1946, the cases of pertussis decreased. After the switch to the aP vaccine in 1996, the amount of cases began to increase. This suggests that the aP vaccine may not have provided immunity that lasts as long as the older wP vaccine.

```
ggplot(cdc) +
  aes(x = Year, y = Number.of.Reported.Pertussis.Cases) +
  geom_point() +
  geom_line() +
```

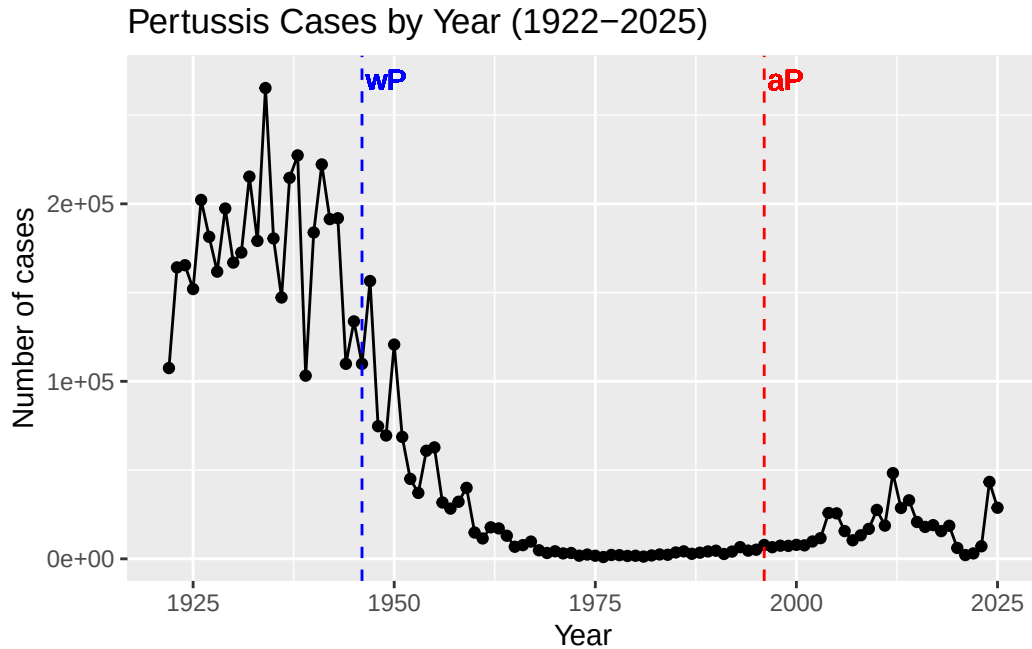
```

geom_vline(xintercept = 1946, color = "blue", linetype = "dashed") +
geom_vline(xintercept = 1996, color = "red", linetype = "dashed") +
geom_text(
  aes(x = 1946, y = 270000, label = "wP"),
  color = "blue",
  hjust = -0.1
) +
geom_text(
  aes(x = 1996, y = 270000, label = "aP"),
  color = "red",
  hjust = -0.1
) +
labs(
  title = "Pertussis Cases by Year (1922-2025)",
  x = "Year",
  y = "Number of cases"
)

```

Warning in geom_text(aes(x = 1946, y = 270000, label = "wP"), color = "blue", : All aesthetic mappings should be placed in aes(). Please consider using `annotate()` or provide this layer with data containing a single row.

Warning in geom_text(aes(x = 1996, y = 270000, label = "aP"), color = "red", : All aesthetic mappings should be placed in aes(). Please consider using `annotate()` or provide this layer with data containing a single row.



Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

After the introduction of the aP vaccine, pertussis cases began to gradually increase in the U.S. This meant that the aP vaccine causes fewer side effects, but doesn't provide longer immunity. Other reasons can include improved PCR testing and bacterial evolution that help *B. pertussis* avoid vaccine induced immunity.

Exploring CMI-PB Data

```
# Allows us to read, write and process JSON data
library(jsonlite)
subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)
head(subject, 3)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	2	wP	Female	Not Hispanic or Latino	White
3	3	wP	Female	Unknown	White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset

```
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
```

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

There are 87 aP vaccinated subjects and 85 wP vaccinated subjects.

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q5. How many Male and Female subjects/patients are in the dataset?

There are 112 female and 60 male patients in the dataset.

```
table(subject$biological_sex)
```

```
Female    Male
   112     60
```

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc...)?

```
table(subject$race, subject$biological_sex)
```

	Female	Male
American Indian/Alaska Native	0	1
Asian	32	12
Black or African American	2	3
More Than One Race	15	4
Native Hawaiian or Other Pacific Islander	1	1
Unknown or Not Reported	14	7
White	48	32

```
library(lubridate)
```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

date, intersect, setdiff, union

```
today()
```

```
[1] "2026-06-01"
```

```
today() - ymd("2000-01-01")
```

Time difference of 9648 days

```
library(jsonlite)
library(lubridate)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(ggplot2)

subject$age <- today() - ymd(subject$year_of_birth)

ap <- subject %>% filter(infancy_vac == "aP")
wp <- subject %>% filter(infancy_vac == "wP")
```

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

The average age of wP individuals is 28 years. The average age of aP individuals is 37 years. They are significantly different.

```
x <- t.test(
  time_length(wp$age, "years"),
  time_length(ap$age, "years")
)
```

```
x$p.value
```

```
[1] 2.372101e-23
```

```
subject <- read_json(
  "https://www.cmi-pb.org/api/subject",
  simplifyVector = TRUE
)

subject$age <- today() - ymd(subject$year_of_birth)
```

```
library(dplyr)

ap <- subject %>% filter(infancy_vac == "aP")

round( summary( time_length( ap$age, "years" ) ) )
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	27	28	28	29	35

```
wp <- subject %>% filter(infancy_vac == "wP")

round(summary(time_length(wp$age, "years")))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	33	35	37	40	58

Q8. Determine the age of all individuals at time of boost?

```
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

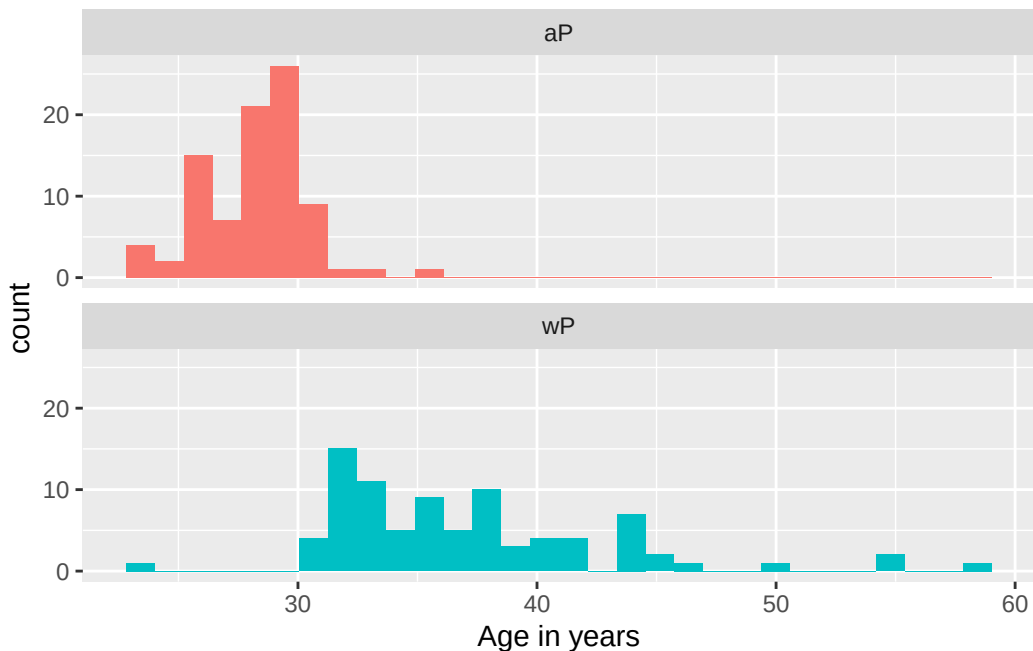
```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

Q9. With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?

These two groups are significantly different. The histogram shows that the aP group is mostly younger around mid-20s to early 30s. The wP group is older and has a wider age range, going up to 60 years old. The t-test gave a p-value of 2.37×10^{-23} , which is smaller than 0.05. This means that the difference in age between the aP and wP groups are statistically significant. This is important because age can affect immune responses, so age is a factor when you compare the two vaccines.

```
ggplot(subject) +  
  aes(time_length(age, "year"),  
       fill=as.factor(infancy_vac)) +  
  geom_histogram(show.legend=FALSE) +  
  facet_wrap(vars(infancy_vac), nrow=2) +  
  xlab("Age in years")
```

``stat_bin()` using `bins = 30`. Pick better value `binwidth`.`



```
# Or use wilcox.test()  
x <- t.test(time_length( wp$age, "years" ),  
            time_length( ap$age, "years" ))
```

```
x$p.value
```

```
[1] 2.372101e-23
```

```
# Complete the API URLs...
library(jsonlite)
library(dplyr)

subject <- read_json(
  "https://www.cmi-pb.org/api/subject",
  simplifyVector = TRUE
)

specimen <- read_json(
  "https://www.cmi-pb.org/api/specimen",
  simplifyVector = TRUE
)

titer <- read_json(
  "https://www.cmi-pb.org/api/plasma_ab_titer",
  simplifyVector = TRUE
)
```

Q9. Complete the code to join `specimen` and `subject` tables to make a new merged data frame containing all specimen records along with their associated subject details:

```
meta <- inner_join(specimen, subject)
```

Joining with ``by = join_by(subject_id)``

```
dim(meta)
```

```
[1] 1503  13
```

```
head(meta)
```

	specimen_id	subject_id	actual_day_relative_to_boost			
1	1	1	-3			
2	2	1	1			
3	3	1	3			
4	4	1	7			
5	5	1	11			
6	6	1	32			

	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex
1	0	Blood	1	wP	Female
2	1	Blood	2	wP	Female
3	3	Blood	3	wP	Female
4	7	Blood	4	wP	Female
5	14	Blood	5	wP	Female
6	30	Blood	6	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
5	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
6	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset

Q10. Now using the same procedure join `meta` with `titer` data so we can further analyze this data in terms of time of visit aP/wP, male/female

```
abdata <- inner_join(titer, meta)
```

Joining with ``by = join_by(specimen_id)``

```
dim(abdata)
```

```
[1] 52576    20
```

Q11. How many specimens (i.e. entries in `abdata`) do we have for each `isotype`?

The dataset contains 6,698 IgE entries, 3,240 IgG entries, and 7,968 entries each for IgG1, IgG2, IgG3, and IgG4.

```
table(abdata$isotype)
```

```

IgE   IgG  IgG1  IgG2  IgG3  IgG4
6698 5389 10117 10124 10124 10124

```

Q12. What are the different `$dataset` values in `abdata` and what do you notice about the number of rows for the most “recent” dataset?

I used `table(abdata$dataset)` to count the number of rows from each dataset. The dataset column shows that the data come from multiple CMI-PB dataset releases. The most recent dataset has fewer rows compared to the older datasets. This suggests that it may be incomplete or still ongoing.

```
table(abdata$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
      31520       8085       7301       5670
```

```
unique(abdata$dataset)
```

```
[1] "2020_dataset" "2021_dataset" "2022_dataset" "2023_dataset"
```

Examine IgG Ab Titer Levels

```
igg <- abdata %>% filter(isotype == "IgG")
head(igg)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgG	TRUE	PT	68.56614	3.736992
2	1	IgG	TRUE	PRN	332.12718	2.602350
3	1	IgG	TRUE	FHA	1887.12263	34.050956
4	19	IgG	TRUE	PT	20.11607	1.096366
5	19	IgG	TRUE	PRN	976.67419	7.652635
6	19	IgG	TRUE	FHA	60.76626	1.096457

	unit	lower_limit_of_detection	subject_id	actual_day_relative_to_boost
1	IU/ML	0.530000	1	-3
2	IU/ML	6.205949	1	-3
3	IU/ML	4.679535	1	-3
4	IU/ML	0.530000	3	-3
5	IU/ML	6.205949	3	-3
6	IU/ML	4.679535	3	-3

	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex
1	0	Blood	1	wP	Female

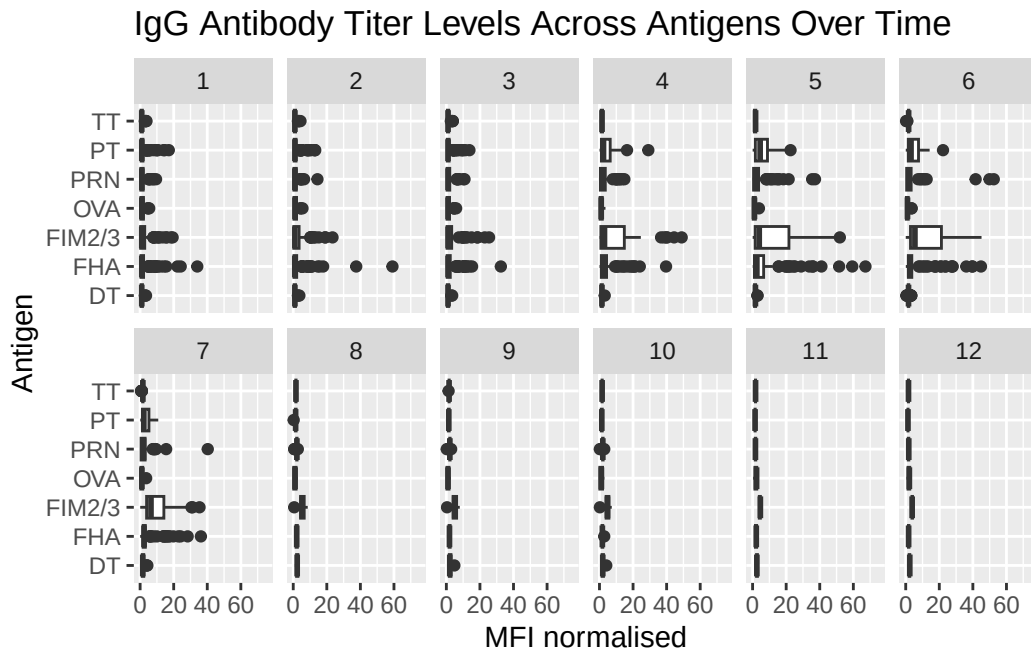
2		0	Blood	1	wP	Female
3		0	Blood	1	wP	Female
4		0	Blood	1	wP	Female
5		0	Blood	1	wP	Female
6		0	Blood	1	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Unknown	White	1983-01-01	2016-10-10	2020_dataset
5	Unknown	White	1983-01-01	2016-10-10	2020_dataset
6	Unknown	White	1983-01-01	2016-10-10	2020_dataset

Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

```
ggplot(igg) +
  aes(x = MFI_normalised, y = antigen) +
  geom_boxplot() +
  xlim(0, 75) +
  facet_wrap(vars(visit), nrow = 2) +
  labs(
    title = "IgG Antibody Titer Levels Across Antigens Over Time",
    x = "MFI normalised",
    y = "Antigen"
  )
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).

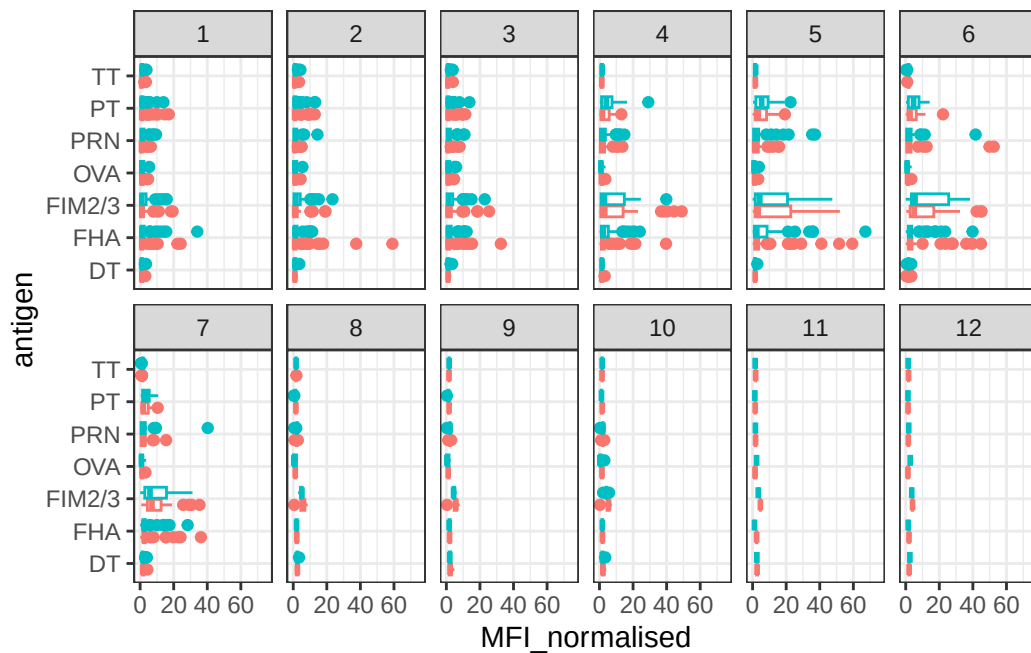


Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

The antigens that show the biggest changes over time are FHA, FIM2/3, PRN, and PT. These are pertussis-related antigens, so the immune system responds to them after the Tdap booster. Ova does not change much because it is not part of the pertussis vaccine and acts more like a control. DT and TT are for diphtheria and tetanus, so their patterns are different from the pertussis antigens.

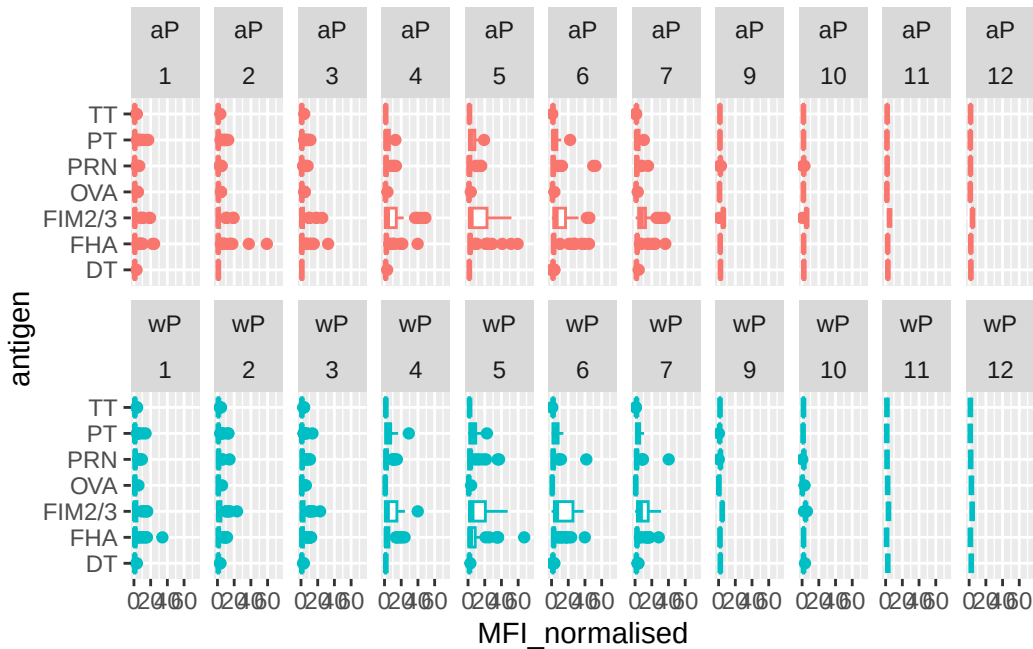
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

Warning: Removed 5 rows containing non-finite outside the scale range (``stat_boxplot()``).



```
igg %>% filter(visit != 8) %>%
ggplot() +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  xlim(0,75) +
  facet_wrap(vars(infancy_vac, visit), nrow=2)
```

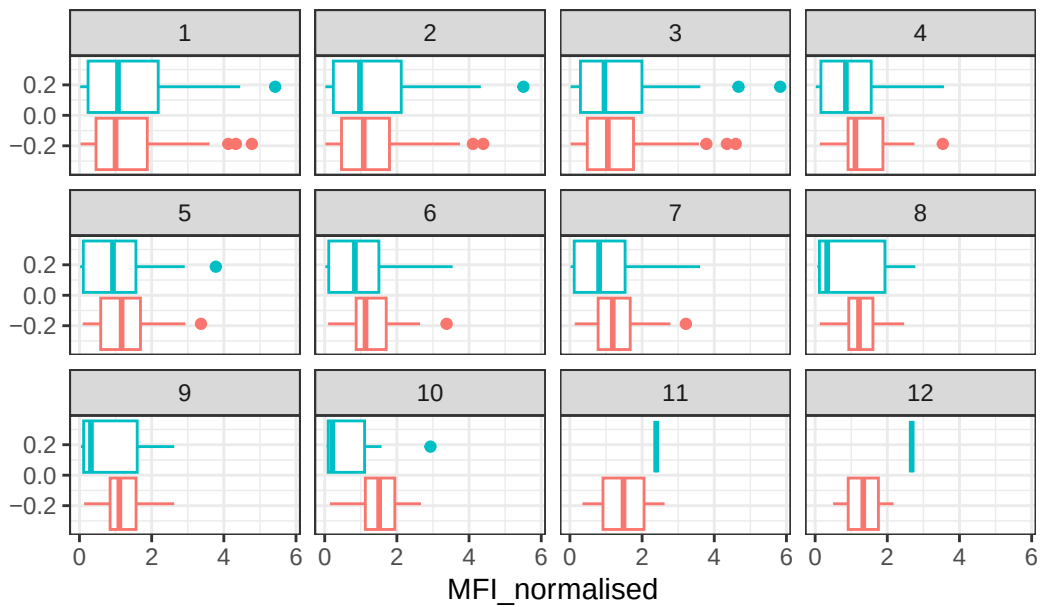
Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).



Q15. Filter to pull out only two specific antigens for analysis and create a boxplot for each. You can chose any you like. Below I picked a “control” antigen (“**OVA**”, that is not in our vaccines) and a clear antigen of interest (“**PT**”, Pertussis Toxin, one of the key virulence factors produced by the bacterium *B. pertussis*).

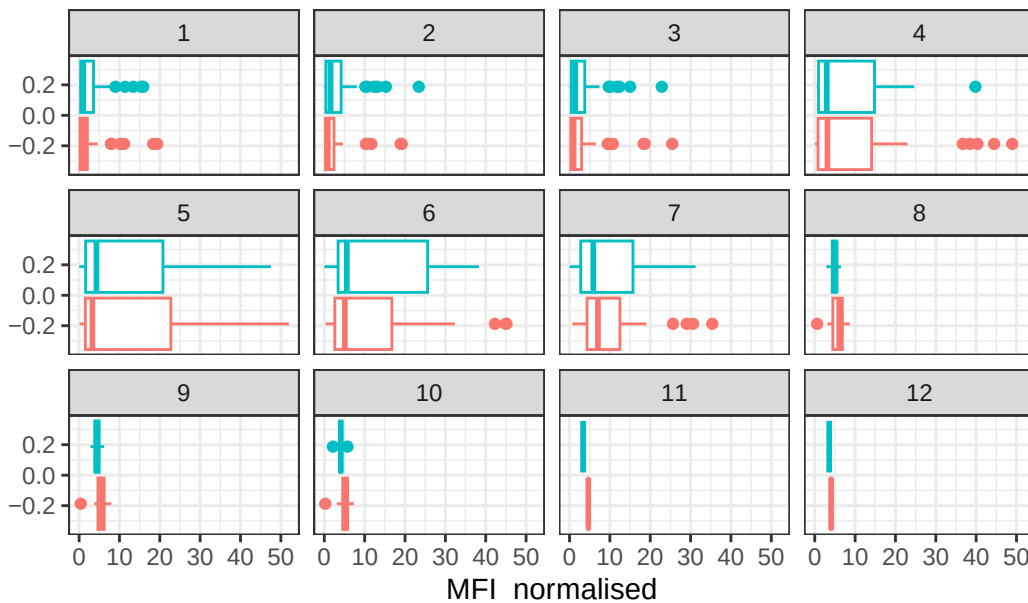
```
filter(igg, antigen == "OVA") %>%
  ggplot() +
  aes(MFI_normalised, col = infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw() +
  labs(title = "OVA antigen levels per visit (aP red, wP teal)"
  )
```

OVA antigen levels per visit (aP red, wP teal)



```
filter(igg, antigen == "FIM2/3") %>%
  ggplot() +
  aes(x = MFI_normalised, col = infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw() +
  labs(title = "PT antigen levels per visit (aP red, wP teal)")
```

PT antigen levels per visit (aP red, wP teal)



Q16. What do you notice about these two antigens time courses and the PT data in particular?

OVA stays low across all visits. This makes sense because it is not part of the vaccine. FIM2/3 shows a stronger response and changes more over time because it is a pertussis-related antigen. For PT, antibody levels increase after the booster, peak around visit 5, and then start to decline. PT levels are much higher than OVA.

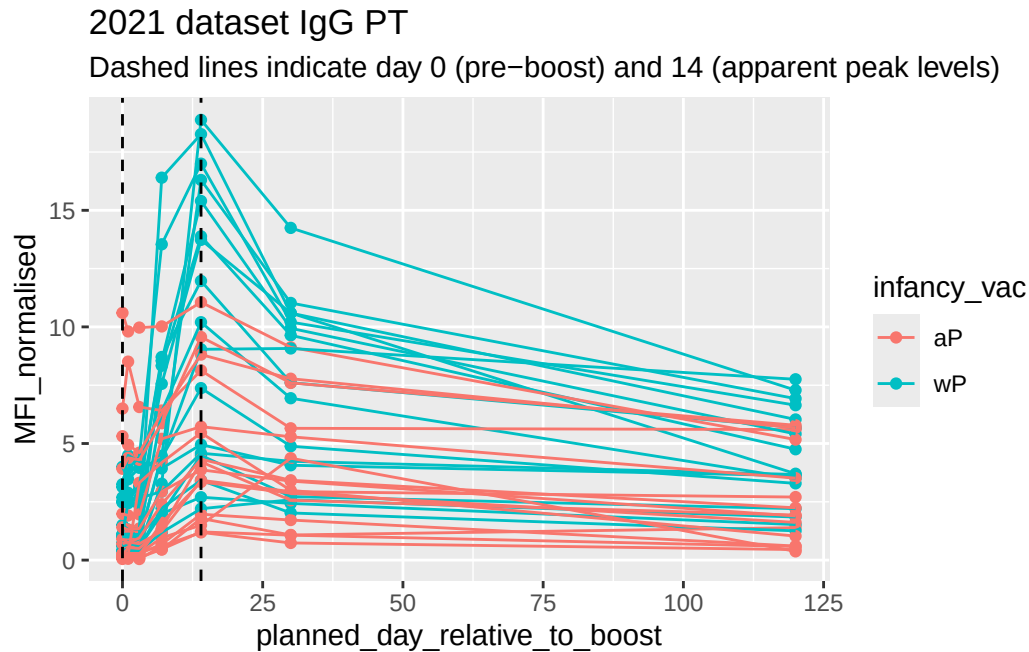
Q17. Do you see any clear difference in aP vs. wP responses?

I do not see a very clear difference between aP and wP responses. Both groups show similar patterns over time.

```
abdata.21 <- abdata %>% filter(dataset == "2021_dataset")

abdata.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
         y=MFI_normalised,
         col=infancy_vac,
         group=subject_id) +
    geom_point() +
    geom_line() +
```

```
geom_vline(xintercept=0, linetype="dashed") +
geom_vline(xintercept=14, linetype="dashed") +
labs(title="2021 dataset IgG PT",
      subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```



Q18. Does this trend look similar for the 2020 dataset?

Yes, the 2020 dataset shows a similar trend. IgG levels against PT increase after the booster, reach a peak around day 14, and then decrease over time. This pattern is similar to the 2021 dataset. This suggest that the PT antibody response rises shortly after boosting and gradually declines.

Obtaining CMI-PB RNASeq Data

```
url <- "https://www.cmi-pb.org/api/v2/rnaseq?versioned_ensembl_gene_id=eq.ENSOG00000211896.7"

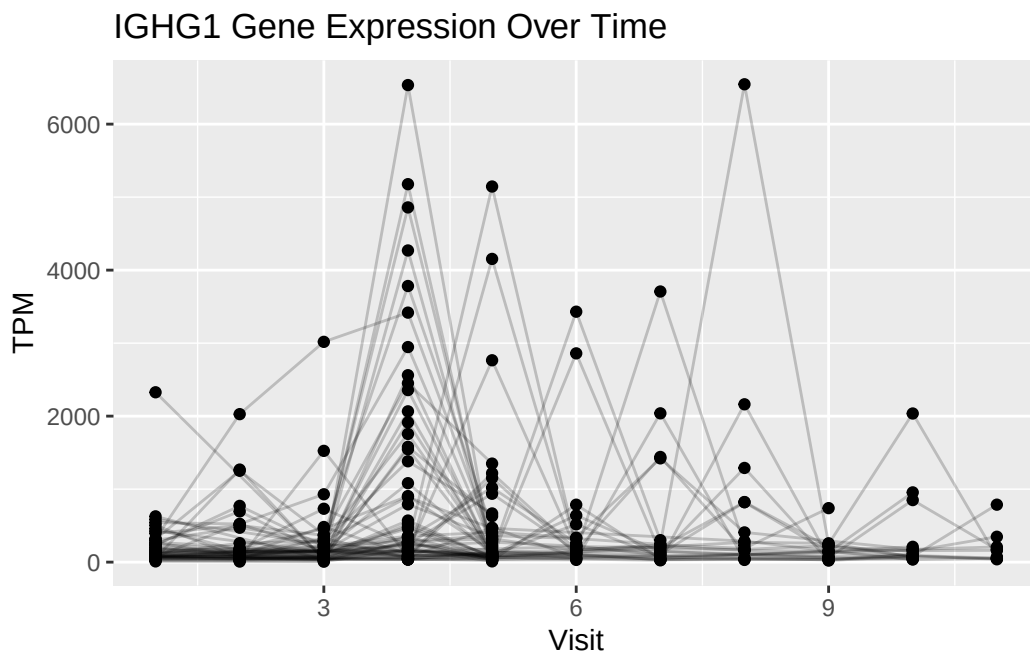
rna <- read_json(url, simplifyVector = TRUE)

#meta <- inner_join(specimen, subject)
ssrna <- inner_join(rna, meta)
```

Joining with ``by = join_by(specimen_id)``

Q19. Make a plot of the time course of gene expression for IGHG1 gene (i.e. a plot of visit vs. tpm).

```
ggplot(ssrna) +  
  aes(x = visit, y = tpm, group = subject_id) +  
  geom_point() +  
  geom_line(alpha = 0.2) +  
  labs(  
    title = "IGHG1 Gene Expression Over Time",  
    x = "Visit",  
    y = "TPM"  
  )
```



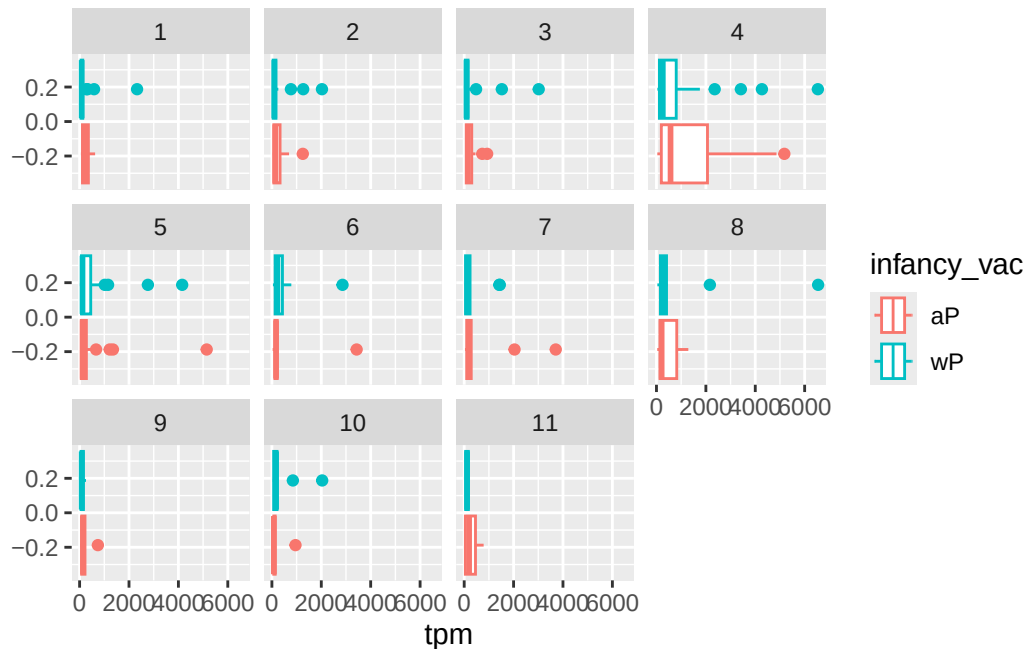
Q20. What do you notice about the expression of this gene (i.e. when is it at it's maximum level)?

IGHG1 expression is highest around visit 4. The expression starts low, then strongly increases at visit 4, before decreasing again at later visits.

Q21. Does this pattern in time match the trend of antibody titer data? If not, why not?

The pattern does not match the antibody titer data. IGHG1 gene expression peaks earlier, around visit 4, because this shows when cells are actively producing antibody-related RNA. Antibody titers can stay high longer because antibodies are proteins that remain in the blood after the gene expression response decreases. RNA expression rises and falls faster than antibody levels.

```
ggplot(ssrna) +
  aes(tpm, col=infancy_vac) +
  geom_boxplot() +
  facet_wrap(vars(visit))
```



```
ssrna %>%
  filter(visit==4) %>%
  ggplot() +
  aes(tpm, col=infancy_vac) + geom_density() +
  geom_rug()
```

